

- Q.31 How will you select a bleaching agent for pulp bleaching?
- Q.32 Discuss the effect of temperature and PH on pulp bleaching?
- Q.33 Explain the alkaline extraction used in the bleaching process?
- Q.34 What is bleach ability? How it is measured?
- Q.35 How pulp strength is affected by different bleaching chemicals.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain the different types of mixers used for carrying out pulp bleaching?
- Q.37 Discuss the multistage bleaching process with the help of neat flow diagram in detail?
- Q.38 Describe the process of pulp bleaching using ozone with the help of neat diagram.

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Roll No.

3rd Sem / Chem, P & P Subject:- Pulp Bleaching

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 What is brightness level of unbleached Kraft pulps?
- a) 10%ISO b) 20%ISO
c) 30%ISO d) 40%ISO
- Q.2 What is the chemical formula of hydrogen peroxide?
- a) HO₂ b) H₂O
c) HO d) H₂O₂
- Q.3 What is the unit of brightness?
- a) Kg b) C
c) %age d) None
- Q.4 What is color of pulp after bleaching?
- a) White b) Pink
c) Blue d) Black
- Q.5 Apparatus for measuring brightness
- a) Micrometer b) Photometer
c) Radiometer d) None

- Q.6 Degree of delignification is measured by _____
 a) Kappa No. b) KMnO_4 no.
 c) Both a & b d) None
- Q.7 Choose any one bleaching process.
 a) O_3 b) CO_2
 c) H_2SO_4 d) CaCl_2
- Q.8 What is made when lignin reacts with NaOH?
 a) $\text{NaOH} + \text{H}_2\text{O}$ b) $\text{Na} + \text{H}_2\text{O}$
 c) Hydro lignin d) Soda lignin
- Q.9 Is chlorine lighter than air?
 a) Yes b) No
- Q.10 What is the full form OXE?
 a) Oxidizing Element
 b) Oxidizing Electron
 c) Oxidizing Equivalent
 d) Oxidizing Electrovalent

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Why alkali used in pulp bleaching operation?
 Q.12 Define mixing.
 Q.13 What is full form of TCF in pulp bleaching?
 Q.14 What is bleaching agent?
 Q.15 Which type of reactions is involved in lignin degrading?

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- Q.16 What is the main advantage of using chlorine dioxide as bleaching agent?
 Q.17 Write the name of a bleaching agent used for high consistency bleaching?
 Q.18 Define brightness.
 Q.19 What is the main objective of later stages in the multistage bleaching?
 Q.20 Write the name of one bleaching chemical.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain the principle of bleaching.
 Q.22 Write five advantages of using multistage bleaching process over single stage bleaching process.
 Q.23 What is post color number.
 Q.24 Explain the term available chlorine.
 Q.25 Write note on peroxide bleaching.
 Q.26 Give advantages and disadvantages of single stage bleaching over.
 Q.27 Differentiate between total chlorine free and elemental chlorine free.
 Q.28 Draw a neat sketch of up flow chloric tower and explain.
 Q.29 Write effect of temperature on chlorination.
 Q.30 Write chlorine water system.

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